

Grade 6 Accelerated
Unit 13
Lesson 1 Homework

Name Answer Key
Date _____
Period _____

1. What is the value of the expression $7 \cdot 4 - (x + 4)$ if $x = 2$?

$$\begin{aligned} &7 \cdot 4 - (2 + 4) \\ &7 \cdot 4 - 6 \\ &28 - 6 \\ &= 22 \end{aligned}$$

2. Find the value of $8 - 3(4 - y) + 2y \cdot 6$ when $y = 4$.

$$\begin{aligned} &8 - 3(4 - 4) + 2(4) \cdot 6 \\ &8 - 3(0) + 2(4) \cdot 6 \\ &8 - 0 + 8 \cdot 6 \\ &8 - 0 + 48 \\ &8 + 48 \\ &= 56 \end{aligned}$$

3. Evaluate $3p + \frac{6}{j} - 1$ for $p = 3$ and $j = 8$.

$$\begin{aligned} &3(3) + \frac{6}{8} - 1 \qquad \frac{78}{8} - \frac{8}{8} = \frac{70 \div 2}{8 \div 2} = \frac{35}{4} \text{ or } 8.75 \\ &9 + \frac{6}{8} - 1 \\ &\frac{72}{8} + \frac{6}{8} - 1 \\ &\frac{78}{8} - 1 \end{aligned}$$

4. Evaluate $10 - 20 \div 2f + 11$ if $f = 9$.

$$\begin{aligned} &10 - 20 \div 2(9) + 11 \\ &10 - 20 \div 2 \cdot 9 + 11 \\ &10 - 10 \cdot 9 + 11 \\ &10 - 90 + 11 \\ &-80 + 11 \\ &= -69 \end{aligned}$$

5. Find the value of $7 \cdot (m - 3m + 2)$ when $m = 1$.

$$\begin{aligned} &7 \cdot (1 - 3(1) + 2) \\ &7 \cdot (1 - 3 + 2) \\ &7 \cdot (-2 + 2) \\ &7 \cdot 0 \\ &= 0 \end{aligned}$$

6. Find the value of $7 - (y + 12)^2 + 6m$ if $m = 4$ and $y = 3$.

$$\begin{aligned} 7 - (3 + 12)^2 + 6 \cdot 4 &= -218 + 24 \\ = 7 - 15^2 + 24 &= -194 \\ = 7 - 225 + 24 \end{aligned}$$


7. Find the value of $r^3 \div x \cdot c$ if $r = 4$, $x = 8$ and $c = 3$.

$$\begin{aligned} 4^3 \div 8 \cdot 3 \\ = 64 \div 8 \cdot 3 &= 24 \\ = 8 \cdot 3 \end{aligned}$$

8. Evaluate the expression $3 \cdot (u^2 - 4) + \frac{k}{2}$ for $u = 2$ and $k = 6$.

$$\begin{aligned} 3 \cdot (2^2 - 4) + \frac{6}{2} &= 0 + 3 \\ = 3 \cdot (4 - 4) + 3 &= 3 \\ = 3 \cdot 0 + 3 \end{aligned}$$


9. What is the value of the expression $5g \cdot h^3 + (h - 1)^2$ if $g = 2$ and $h = 5$?

$$\begin{aligned} 5(2) \cdot 5^3 + (5 - 1)^2 &= 1250 + 16 \\ = 10 \cdot 125 + 4^2 &= 1,266 \\ = 10 \cdot 125 + 16 \end{aligned}$$


10. What is the value of the expression $\left(\frac{b^2-6}{4}\right) \cdot 7 + y^2$ if $b = 4$ and $y = 7$?

$$\begin{aligned} \left(\frac{4^2-6}{4}\right) \cdot 7 + 7^2 &= 2.5 \cdot 7 + 49 \\ = \left(\frac{16-6}{4}\right) \cdot 7 + 49 &= 17.5 + 49 \\ = \frac{10}{4} \cdot 7 + 49 &= 66.5 \end{aligned}$$
